RAFFLES INSTITUTION
Y6 PHYSICS PAPER 4
Higher 2

CANDIDATE
NAME

CLASS	2	6	S	0	2	B

PHYSICS **9478/04**

Paper 4 Practical **2 hour 30 minutes**

You must answer on the question paper.

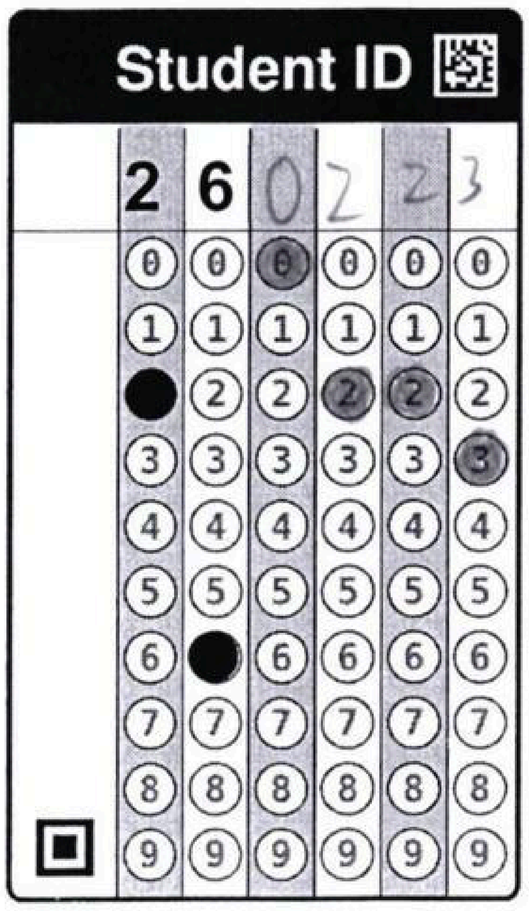
INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name and class in the boxes at the top of the page.
- Shade the lozenges in the Student ID box below according to your RI Exam Index No.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- You may use an approved calculator.
- You may use a spreadsheet to process and analyse data.
- Write the details of the shift and laboratory in the boxes provided.
- You will be allowed to work with the apparatus for a maximum of 1 hour 15 minutes for each section.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [ ].

Student ID



Shift
3
Laboratory
bio lab 2

For Examiner's Use		
1	5	/ 9
2	4	/ 17
3	16	/ 24
Total	25	/ 50

1 In this experiment, you will investigate the principle of moments.

(a) Place the metre rule on the pivot. Adjust the position of the metre rule until it balances as shown in Fig. 1.1. The centre of gravity (c.g.) of the metre rule is above the pivot.

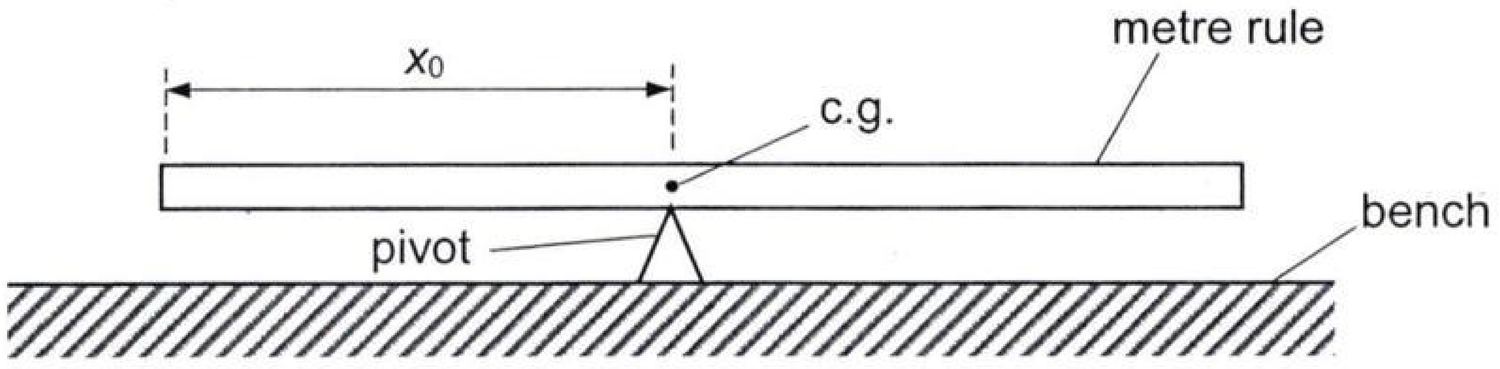


Fig. 1.1

The distance between one end of the metre rule and the pivot is x_0.

Measure and record x_0.

✓ Correct (1)

1 / 1 $X_0 = \frac{50+50}{2} = 50\text{cm} = 0.50\text{m}$ $\frac{49.6+49.}{2}$ $\frac{49.5+49.5}{2} = 49.5\text{ cm} = 0.495\text{ m}$

x_0 = ~~0.500 m~~ 0.495 m [1]

(b) **(i)** Place a mass m of 50 g on the metre rule such that the centre of mass m is approximately 2 cm from the same end of the metre rule as **(a)**. The distance between the end of the metre rule and the centre of mass m is l_0.

Adjust the position of the pivot until the metre rule balances as shown in Fig. 1.2. The distance between the end of the metre rule and the pivot is x.

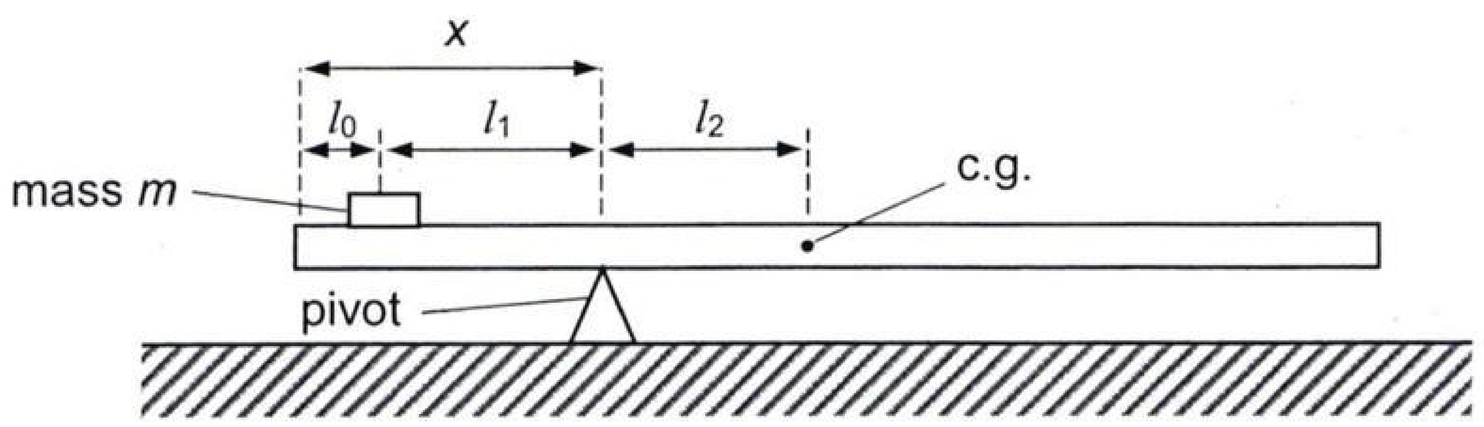


Fig. 1.2

$50\text{g} = 0.05\text{kg}$

Record m, l_0 and x.

$l_0 = \frac{10.0+10.0}{2} = 10.0\text{cm} = 0.100\text{m}$

$l_1 = \frac{27.0+27.0}{2} = 27.0\text{cm} = 0.270\text{m}$

$l_2 = \frac{12.5+12.5}{2} = 12.5\text{cm} = 0.125\text{m}$

$X = l_0 + l_1$
$= 37\text{cm} = 0.370\text{m}$

m = ~~50~~ 0.05 kg

l_0 = 0.100 m

x = 0.370 m

(ii) The distance between the centre of mass m and the pivot is l_1. The distance between the pivot and the c.g. of the metre rule is l_2.

Calculate l_1 and l_2.

1 / 2

$l_1 = X - l_0$
$= 0.370 - 0.100$
$= 0.270\,m$

l1: wrong dp / unit / calculation (-1)

$l_2 = X_0 - X$
$= 0.495 - 0.370$
$= 0.125\,m$

l_1 = 0.270 m .....

l_2 = 0.125 m ..... [2]

(iii) Vary m by adding masses and repeat **(b)(i)** and **(b)(ii)**.

Present your results clearly.

l_0 const. at 0.100m

1 / 3

m/kg	X/m	l_1/m	l_2/m	
0.050	0.370	0.270	0.125	
0.070	0.339	0.239	0.156	
0.100	0.305	0.205	0.190	
0.120	0.288	0.188	0.207	
0.150	0.266	0.166	0.229	
0.170	0.254	0.154	0.241	

[3]

Wrong / table he

Wr for

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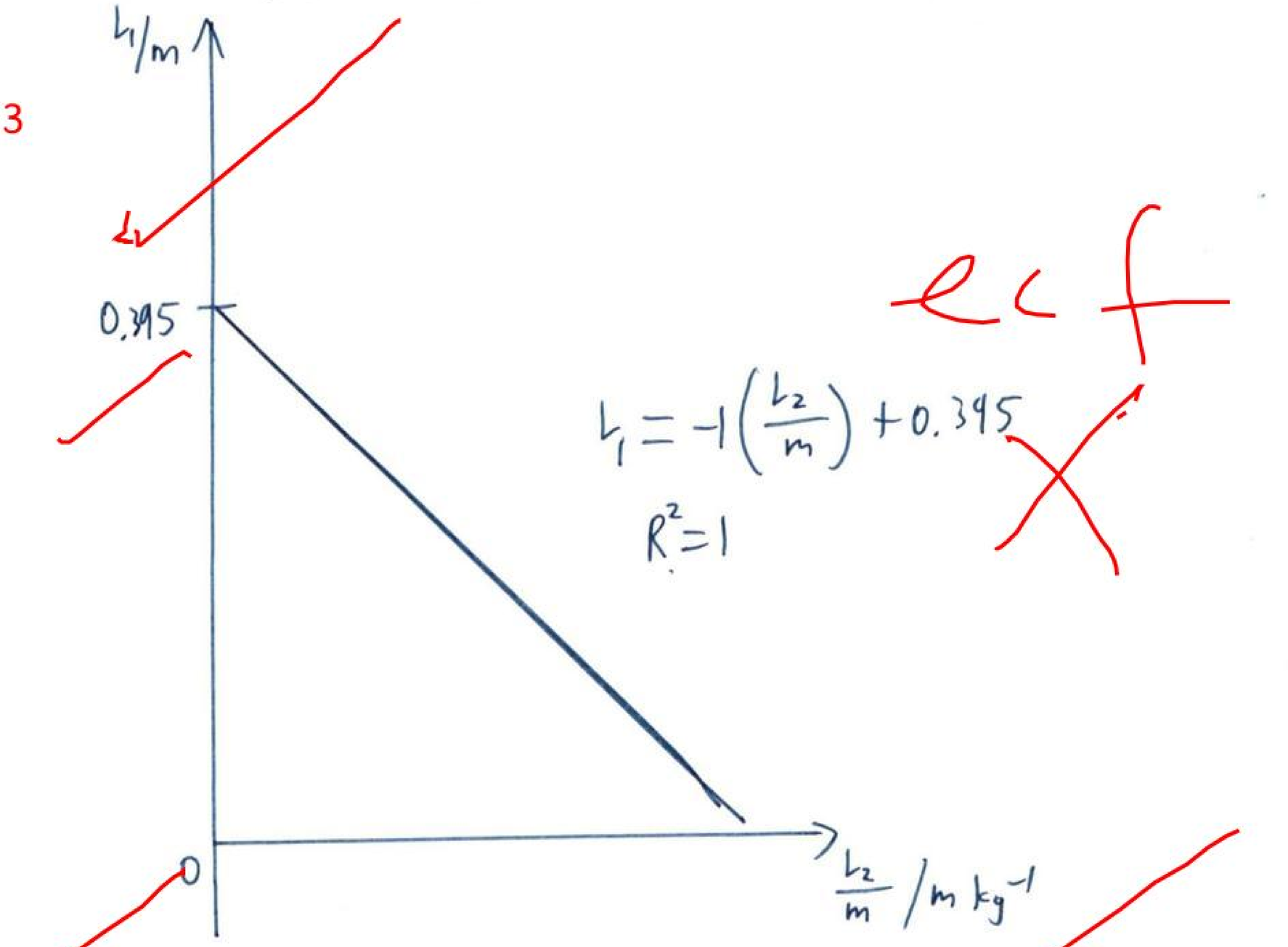
(c) It is suggested that m, l_1 and l_2 are related by the expression:

$$l_1 = \frac{Ml_2}{m}$$

where M is the mass of the metre rule.

Plot a suitable graph using a spreadsheet to determine M.

Sketch the graph in the space below. Write down the equation of the trendline.



M = ... [3]

[Total: 9]

In this experiment, you will investigate the motion of an oscillating mass.

(a) Set up the apparatus as shown in Fig. 2.1.

The total mass of the load consisting of mass hanger and slotted masses is 300 g.

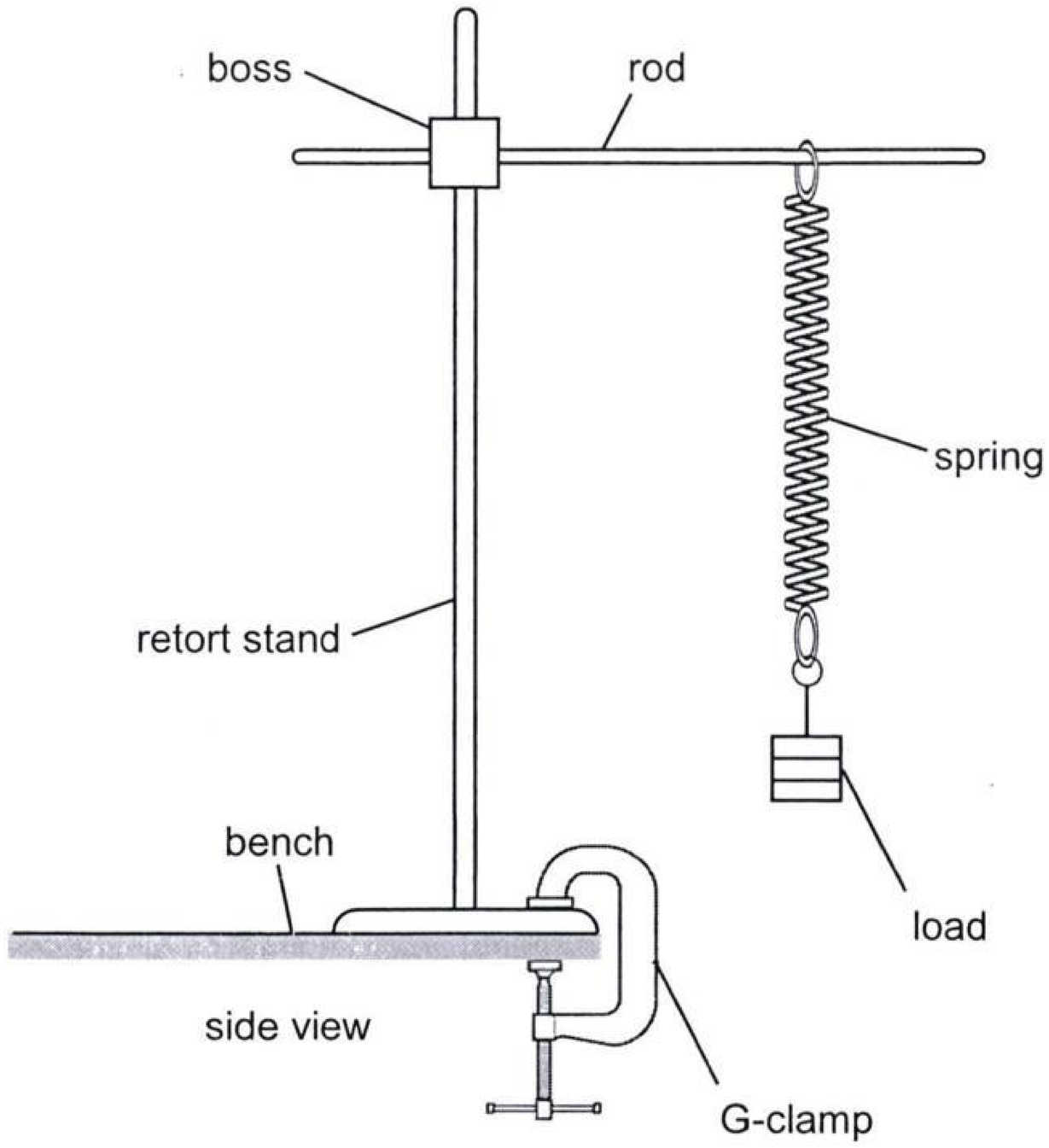


Fig. 2.1

(i) Pull the load down and release it such that the load oscillates vertically.

Determine the period T of the oscillation.

1 / 1

for total time > 20s, oscillation count $N \approx 30$

$T = \frac{(21.55 + 21.55)}{(30)(2)}$ → this is real both readings had exact same timing

$= 0.718333$

$= 0.7183\,s$

T = 0.7183s [1]

(ii) Calculate a length L using the equation:

$$L = \frac{gT^2}{\pi^2}$$

where the acceleration of free fall $g = 9.81\ \text{m s}^{-2}$

1 / 1

$L = \frac{(9.81)(0.7183)^2}{\pi^2}$

$= 0.51284$

$= 0.513\ \text{m}$

L = 0.513m ..... [1]

(b) Set up the apparatus as shown in Fig. 2.2 by attaching a string between the spring and the load.

The distance d is measured from the top of the string to the centre of the load. Adjust the knots of the string so that d is approximately equal to the calculated value of L in **(a)(ii)**.

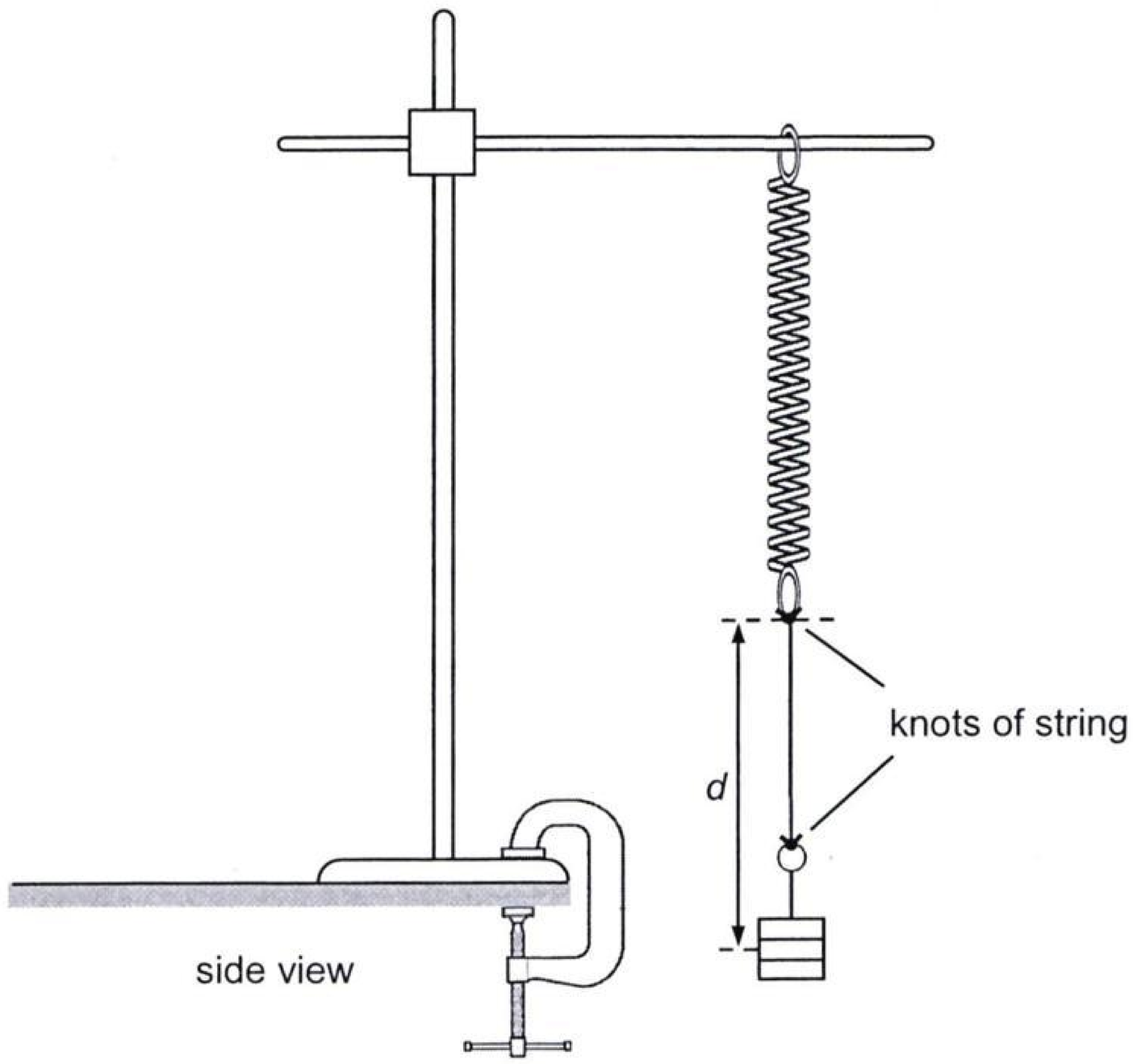


Fig 2.2

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(i) Measure and record d.

0 / 1

$d = \frac{38.6+38.6}{2}$

$= 38.6\text{cm}$

$= 0.386\text{ m}$

$\pm 1\text{cm}$ for L

d = 0.386m ..... [1]

(ii) Displace the load approximately 10 cm to the right and release it. The load will oscillate left and right, as well as up and down. The separation between the coils of the spring increases and decreases.

At certain times, the separation between the coils of the spring appear to remain constant. The time interval T_b between two such consecutive occurrences is known as the beat period. 30.45 30.17 30.27

Determine the beat period T_b of these oscillations.

0 / 1

for total time > 20s,

$T_b = \frac{(30.17+30.27)}{(20)(2)}$ X

$= 1.511\text{ s}$

> 2s

T_b = 1.511s X ..... [1]

(iii) State and explain what you did in **(b)(ii)** to ensure that your value of T_b is reliable.

1 / 1

I took 2 readings and averaged them to calculate for T_b. This is to reduce random error and get more reliable data by taking more data.

..... [1]

6m small pulley — 18min (?)

(c) The beat period in **(b)(ii)** is dependent on the angular displacement θ of the oscillating sys from the vertical and the length d in **(b)(i)** by the equation:

$$T_b = K\theta^p d^q$$

where θ is measured in radians and K, p and q are constants.

$\ln T_b = \ln K + \ln \theta^p + \ln d^q$
$= \ln K + p \ln \theta + q \ln d$
$\ln K\theta^p d^q$
$= p \ln \theta + \ln K d^q$

Design an experiment to determine the values of p and q.

You should pay particular attention to:

- the equipment you would use
- the procedure to be followed and
- the control of variables

P2 - Meas d with metre rule (1)

1 / 6

1. using the setup from fig. 2.2, measure d using a metre rule

~~and the~~

2. place a metre rule on the table such that it is ~~perpendicular~~ parallel to the wooden rod in the retort stand, placing it 10 cm away perpendicular to the load

towards the rule

3. pull the load ~~to the~~ such that it touches the metre rule

missing eqn for rad conversion

4. measure the angular displacement θ of the load using a protractor.

5. repeat the step in (b). (ii). to obtain T_b

6. ~~repeat step 5 until 6 sets of T_b~~, repeat step 5 changing the distance of the metre rule from the load such that a suitable increment in θ is obtained.

missing constant variable + details

7. repeat step 6 until 6 sets of readings are obtained.

8. tabulate values for $\ln T_b$ and $\ln\theta$ to plot a graph of $\ln\theta$ against $\ln T_b$ and obtain a trendline in the form $\ln T_b = p \ln\theta + \ln K d^q$

9. The y-intercept value of the linear trendline is $\ln K d^q$, and gradient is p.

10. calculate q values of p and q from the y-intercept and gradient values.

11.

[6]

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(d) A student conducts a preliminary investigation of the force–extension relationship of a spr
The student uses a digital data logger connected to a force sensor and a displacement sensor
to record force F and extension x data continuously as the spring is loaded to a maximum extension of 35.0 cm and then unloaded.

The student's data is recorded in the digital file: datasetQ2d.xlsx.

(i) Use the student's data to estimate the extension x_L when the spring stops obeying Hooke's law.

0 / 1

x_L = 0.150 m [1]

(ii) Explain how you obtained the value of x_L.

0 / 1

By looking at where the gradient no longer follows the original, initially constant gradient from X = 0m to X = 0.15m.

What graph?

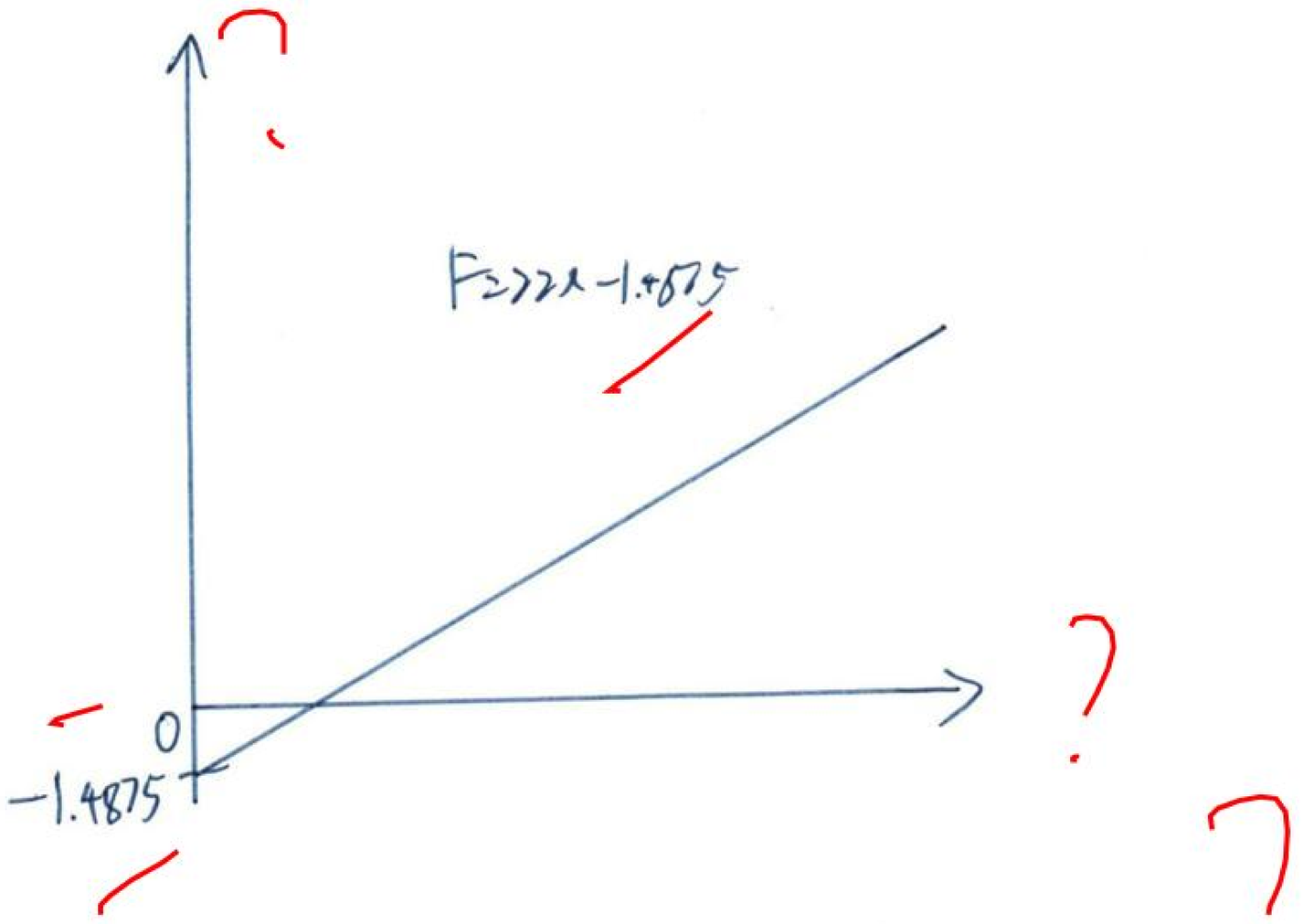
[1]

(iii) Energy is lost in the spring during the loading and unloading cycle. The spring constant k is different during loading and unloading.

1. Plot a suitable graph using a spreadsheet to determine k during unloading.

Sketch the graph in the space below. Write down the equation of the trendline.

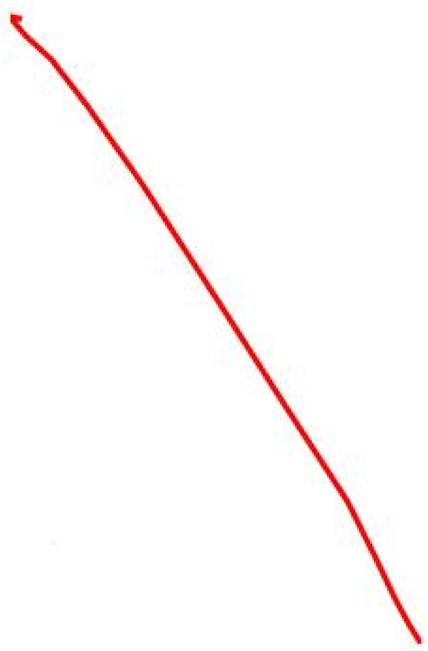
0 / 2



k = ... [2]

2. Calculate the total energy E that is lost in the spring during the loading and unloading cycle. Explain your working clearly.

0 / 2



E = ... [2]

[Total: 17]

3 In this experiment, you will investigate an electrical circuit.

(a) (i) Measure and record the length L_x and the diameter d_x of the resistance wire labelled **X**.

1 / 1

$L_x = \frac{50.8+50.8}{2} = 50.8\text{cm}$
$= 0.508\text{m}$

$d_x = \frac{0.236+0.236}{2} = 0.236\text{mm}$
$= 2.36 \times 10^{-4}\text{m}$

L_x = ~~0.0~~ 0.508m

d_x = 2.36×10^{-4} m [1]

(ii) Calculate the cross-sectional area A_x of the resistance wire **X**.

1 / 1

$A_x = \pi \left(\frac{d_x}{2}\right)^2$
$= \pi \left(\frac{2.36 \times 10^{-4}}{2}\right)^2$
$= 4.37 \times 10^{-8}\ \text{m}^2$

A_x = $4.37 \times 10^{-8}\ \text{m}^2$ [1]

(iii) Estimate the percentage uncertainty in your value of A_x.

Smallest graduation of micrometer screw gauge $= 0.001\text{mm} = 1 \times 10^{-6}\text{m}$

0 / 1

$\frac{\Delta A_x}{A_x} = 2\left(\frac{\Delta d_x}{d_x}\right)\left(\frac{1}{2}\right) \times 100\%$
$= 2\left(\frac{3 \times 10^{-6}}{2.36 \times 10^{-4}}\right) \times 100\%$
$= 1.3\%$ (2 s.f.)

percentage uncertainty in A_x = 1.3% [1]

(b) Set up the circuit as shown in Fig. 3.1 using a **two-cell battery**. Connect both ends of resistance wire **X** to the circuit. ST is a resistance wire mounted on a metre rule.

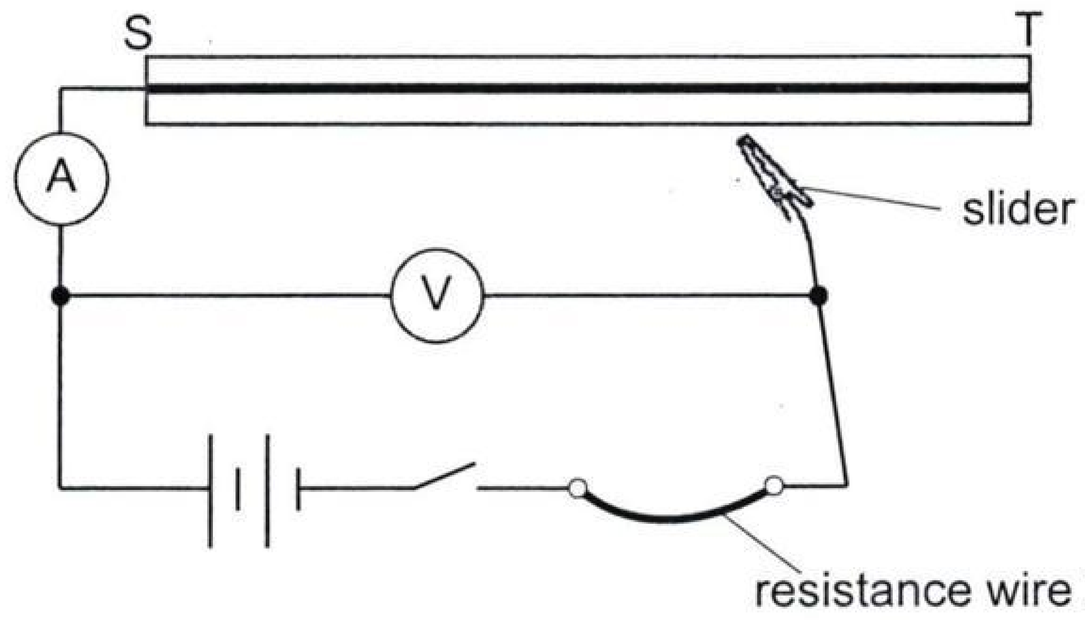


Fig. 3.1

(i) Close the switch. Adjust the position of the slider on the resistance wire ST until the ammeter reading I is lowest. The voltmeter reading is V.

Record I and V.

I = 0.205A

V = 1.367V

(ii) Vary the position of the slider on ST to obtain more readings of I and V.

Present your results clearly. distance is d/m from V lowest value of I position of

3 / 3

I/A	V/V	d/m
0.205	1.367	0.00
0.218	1.247	0.16
0.228	1.146	0.232
0.255	0.991	0.348
0.279	0.789	0.464
0.310	0.560	0.580

[3] 3

(iii) I and V are related by the expression:

$$V = P - QI$$

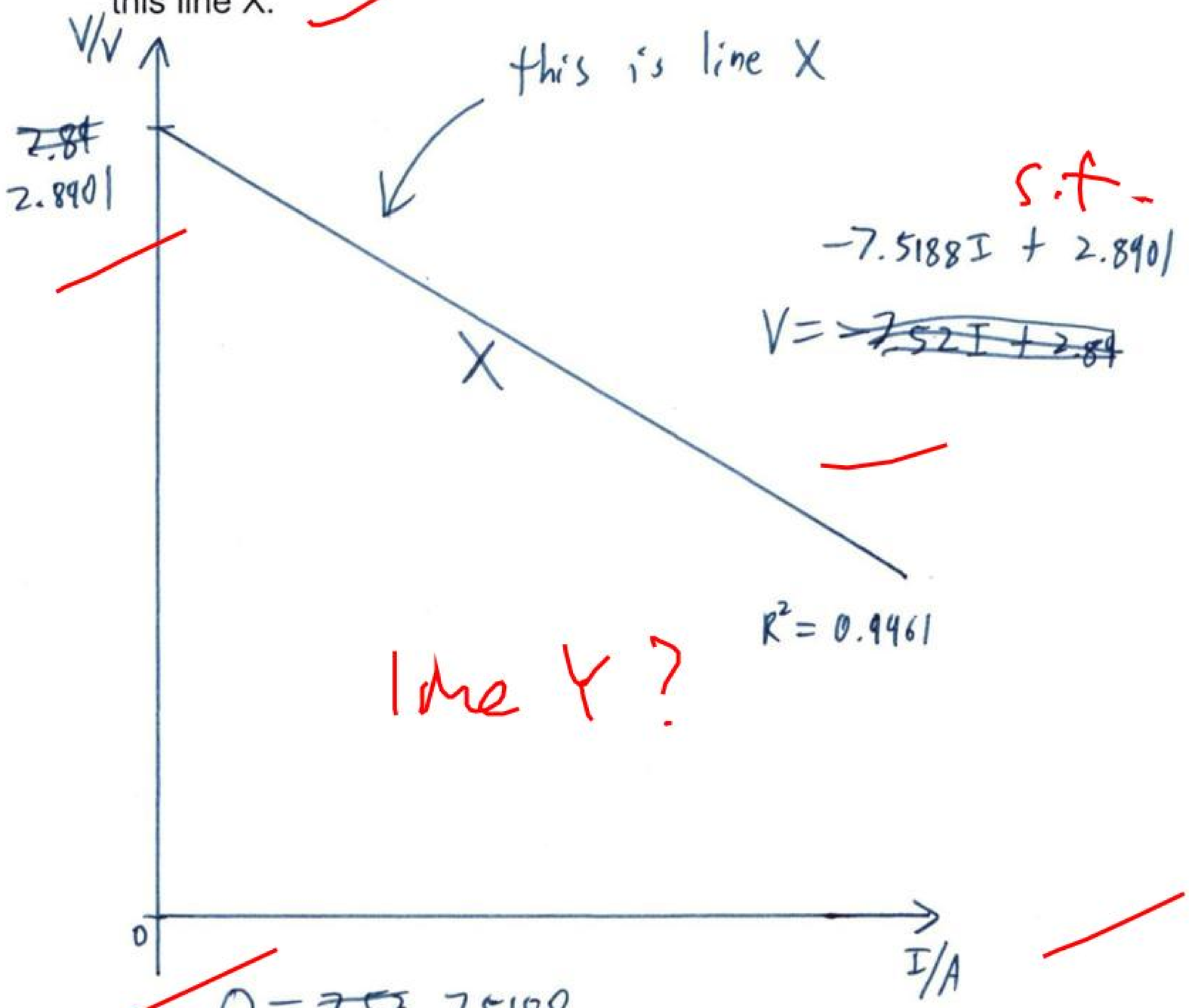
where P and Q are constants.

The resistance per unit length α_x of wire **X** is given by:

$$Q = \alpha_x L_x$$

Plot a suitable graph using a spreadsheet to determine α_x.

Sketch the graph in the space below. Write down the equation of the trendline and label this line X.



$Q_x = 7.52$ ~~7.52~~ 7.5188

$\alpha_x = \frac{Q_x}{L_x} = \frac{7.5188}{0.508} = 14.8$ ~~14.8~~ $\alpha_x =$ 14.8 Ω m^{-1} [3]

(c) (i) Measure and record the length L_Y and the diameter d_Y of the resistance wire labelled **Y**.

$L_Y = \frac{50.6+50.6}{2} = 50.6\text{ cm} = 0.506\text{ m}$

$d_Y = \frac{0.190+0.190}{2} = 0.190\text{ mm} = 1.90 \times 10^{-4}\text{ m}$

$L_Y =$ 0.506 m

$d_Y =$ 1.90×10^{-4} m

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(ii) Calculate the cross-sectional area A_Y of the resistance wire **Y**.

$A_Y = \pi\left(\frac{1.90\times10^{-4}}{2}\right)^2$

$= 2.84\times10^{-8}\ m^2$

$A_Y =$ $2.84\times10^{-8}\ m^2$

(iii) The resistance per unit length of wire **Y** is α_Y.

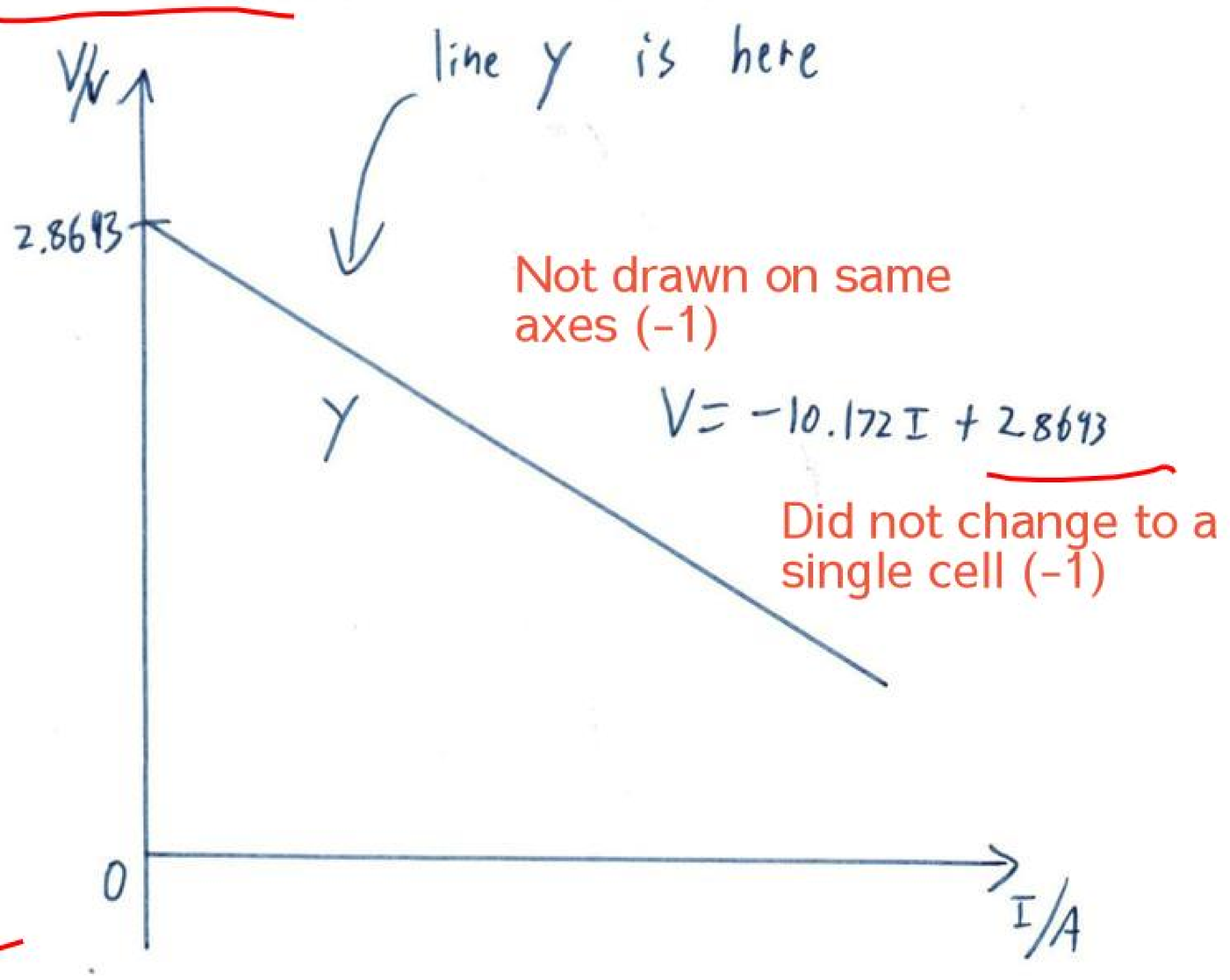
Replace the two-cell battery in Fig. 3.1 with a single cell.

Repeat **(b)(i), (b)(ii) and (b)(iii)** to determine α_Y.

Sketch your graph on the same axes in **(b)(iii)**. Write down the equation of the trendline and label this line Y.

1 / 3

l/m	V/V	I/A
0.00	1.102	0.174
0.16	0.985	0.184
0.32	0.859	0.196
0.48	0.799	0.206
0.64	0.591	0.226
0.80	0.338	0.247



$Q_y = 10.172$

$a_y = \frac{Q_y}{L_y} = \frac{10.172}{0.506} = 20.1\ \Omega\ m^{-1}$

$\alpha_Y =$ $20.1\ \Omega\ m^{-1}$ [3]

(d) It is suggested that the resistance per unit length α of a wire is given by the relationship:

$$A = \frac{k}{\alpha}$$

where A is the cross-sectional area and k is a constant.

(i) Use your values from **(a)(ii)**, **(b)(iii)**, **(c)(ii)** and **(c)(iii)** to determine two values of k.

1 / 1

$A_x = \frac{k_1}{\alpha_x}$ $k = A\alpha$

$k_1 = A_x \alpha_x$

$= (4.37 \times 10^{-8})(14.8)$

$= 6.4676 \times 10^{-7}$

$= 6.47 \ \Omega m$ 1

$k_2 = A_y \alpha_y$

$= (2.84 \times 10^{-8})(20.1)$

$= 5.7084 \times 10^{-7}$

$= 5.71 \ \Omega m$ 1

first value of k = 6.47 Ωm $\times 10^{-7}$

second value of k = 5.71 Ωm [1] 0

(ii) State whether the results of your experiment support the suggested relationship.

0 / 1

Justify your conclusion by referring to your value in **(a)(iii)**.

Plotting a graph of I against V gives a linear graph with R^2 value close to 1, which suggests a linear relationship. This follows the linear relationship of $V = \rho - QI$, where the plotted graph gave a linear equation $V = 2.8901 - 7.5188I$. [1] 0

(e) Suggest a significant source of error in this experiment.

0 / 2

Suggest an improvement that could be made to the experiment to address this error. You may suggest the use of other apparatus or a different procedure.

source of error Poor contact with slider

improvement Use a

[2]

(f) The core of a pencil is made of graphite, which is an electrical conductor of resistivity ρ. A student wishes to determine the thickness t of a layer of pencil shading using electrical method.

Half of a strip of paper is **uniformly and heavily shaded** with a pencil as shown in Fig. 3.2.

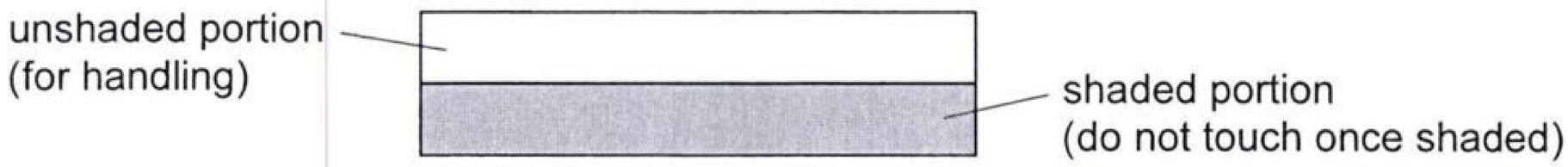


Fig. 3.2

The shaded portion of the paper is used in a circuit with a **two-cell battery** of electromotive force (e.m.f.) E and a fixed resistor R_f as shown in Fig. 3.3.

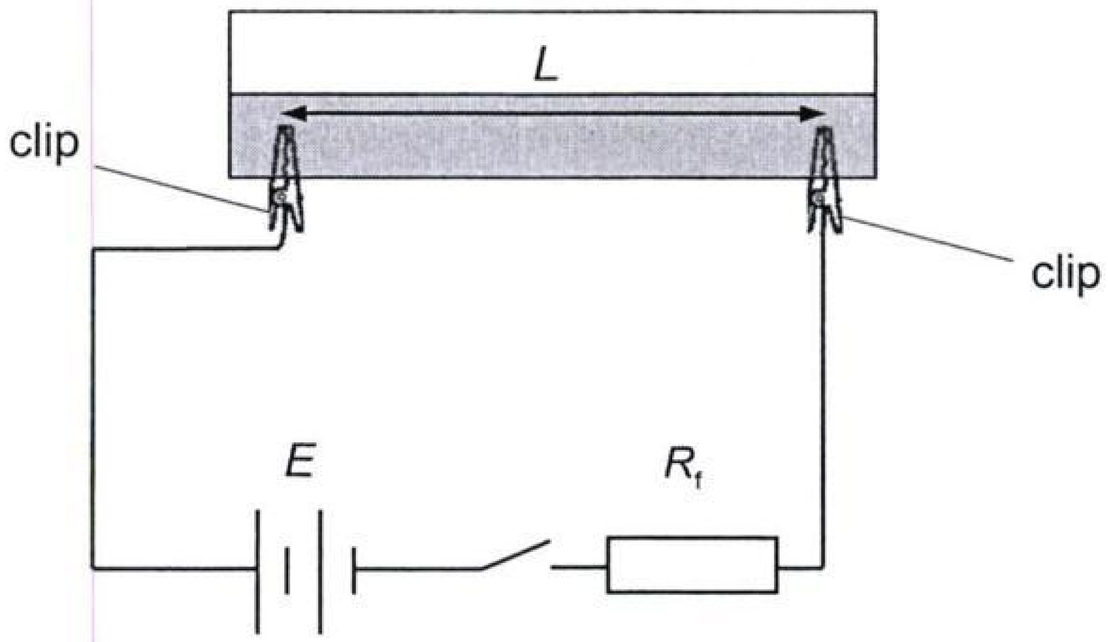


Fig. 3.3

(i) Prepare the strip of pencil shading using the paper and pencil provided and set up the circuit in Fig. 3.3.

1 / 2

Measure and record the distance L between the clips.

Measure and record the e.m.f. E and the potential difference (p.d.) V between the clips.

$L = \frac{8.0+8.i}{2} = 8.0\text{cm} = 0.08\text{m}$

$L =$ 0.08m d.p.

$E =$ 2.93V

$V =$ 0.717V [2]

(ii) Theory suggests that:

1 / 1

$$t = 1.0 \times 10^{-6} LE\left(\frac{1}{V} - \frac{1}{E}\right)$$

Calculate the thickness t of the pencil shading.

$t = (1\times10^{-6})(0.08)(2.93)\left(\frac{1}{0.717} - \frac{1}{2.93}\right)$

$= 2.4691\times10^{-7}$

$= 2.47\times10^{-7}$ m

$t =$ 2.47×10^{-7} m [1]

5m small planning — 15 min

(iii) The p.d. V between the clips on the strip of pencil shading is related to the distance between the clips and the e.m.f. E of the battery by the expression:

$$\frac{1}{V} = \frac{\beta}{L} + \frac{1}{E}$$

where β is a constant.

Plan an investigation to determine the value of β.

Include in your plan:

- the equipment you would use
- the procedure to be followed
- any precautions that should be taken to improve the accuracy of the experiment.

4 / 5

1. set up a circuit ~~connecting~~ as shown in fig. 3.3, with a 30cm rule below the shaded portion of the paper. ~~and a~~
2. measure e.m.f. E of battery using voltmeter to obtain $\frac{1}{E}$
3. connect ~~with~~ voltmeter in parallel to the clips on the paper
4. ~~to~~ unclip one side of the clips on the paper ~~to~~ to slide on the paper
5. take readings for V and L while ~~incrementing~~ sliding the clip down by 1cm for each reading
6. repeat ~~for~~ step 5 until 6 readings are obtained
7. tabulate table of $\frac{1}{V}$ and $\frac{1}{L}$.
8. plot a graph of $\frac{1}{L}$ against $\frac{1}{V}$ to obtain a trendline.
9. gradient of trendline is value β.
10. To ~~ensure~~ reduce systematic error, compare measured $\frac{1}{E}$ to the y-intercept of the graph ~~and~~
11. To reduce random error and improve reliability, repeat experiment.

Ruler for measuring L? state clearly (BOP)

Precaution too generic

[5]

[Total: 24]

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